Minimize $z = 4x_1 + x_2 + MR_1 + MR_2$

subject to $3x_1 + x_2 + R_1 = 3$

$4x_1 + 3x_2 - x_3 + R_2 = 6$

$x_1 + 2x_2 + x_4 = 4$

$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0$

!

	Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z-row	z	−4	−1	0	−M → 0	−M → 0	0	0
R_1-row	R_1	3	1	0	1	0	0	3
R_2-row	R_2	4	3	−1	0	1	0	6
x_4-row	x_4	1	2	0	0	0	1	4

$\rightarrow 9M = z_0.$

$(R_1, R_2, x_4) = (3, 6, 4)$

$(R_1\text{-row}) \times (M) + \text{old } z\text{-row} = \text{New } z\text{-row}.$

$(R_2\text{-row}) \times (M) + \text{old } z\text{-row} = \text{New } z\text{-row}.$

$(R_1)M \Rightarrow$	3M	M	0	M	0	0	3M
$(R_2)M \Rightarrow$	4M	3M	−M	0	M	0	6M
old z-	−4	−1	0	−M	−M	0	0
+	7M−4	4M−1	−M	0	0	0	9M.

+? the most positive.

Basic ✓

Basic	x_1 (+ E.V.)	x_2 (+)	x_3 (−)	R_1	R_2	x_4	Solution	min Ratio
min z	7M−4	4M−1	−M	0	0	0	9M $= z_0$ ✓	
L.V. R_1	3	1	0	1	0	0	3	3/3
R_2	4 → 0	3	−1	0	1	0	6	6/4
x_4	1 → 0	2	0	0	0	1	4	4/1

		+E.V.						min ratio
min z	0	$\frac{5}{3}M + \frac{1}{3}$	−M	$\frac{-7}{3}M + \frac{4}{3}$	0	0	$4 + 2M = z_1$	
x_1	1	1/3	0	1/3	0	0	1	$1/1/3 = 3$
L.V. R_2	0	5/3	−1	−4/3	1	0	2	$2/5/3 = 6/5$
x_4	0	5/3	0	−1/3	0	1	3	$3/5/3 = 9/5$

$3 - \frac{4}{3}$

$2\frac{1}{3}$ $-4(x_1 \text{ row}) + R_2^{\text{old}}\text{row} = \text{New } R_2\text{-row}.$

x_1
x_2
x_4

$7M - 4 \rightarrow 0.$

$+ 4 - 7M$

$0.$

$$(4-7M)\cdot\left(\begin{matrix}\text{New}\\ x_1\text{-row}\end{matrix}\right) + \begin{matrix}\text{old}\\ z\text{-row}\end{matrix} = \begin{matrix}\text{New}\\ z\text{-row}\end{matrix}$$

$4-7M$	$\frac{4-7M}{3}$	0	$\frac{4-7M}{3}$	0	0	$4-7M$
$+\ 7M-4$	$4M-1$	$-M$	0	0	0	$9M$
0^{*}	$\frac{5}{3}M+\frac{1}{3}$	$-M$	$-\frac{7}{3}M+\frac{4}{3}$	0	0	$4+2M$
	$\frac{-7}{3}+4$					

Big-M method

If all artificial variables are equal to zero in the optimal solution, then we have found the optimal solution to the original problem.
If any artificial variables are positive in the optimal solution, then the original problem is infeasible.

Example 1:

min $z = -3x_1+4x_2$
s.t. $x_1+x_2 \le 4$
$2x_1+3x_2 \ge 18$
$x_1, x_2 \ge 0$

$\Rightarrow$ min $z = -3x_1 + 4x_2 + M.R_1$
s.t. $x_1 + x_2 + S_1 = 4$
$2x_1 + 3x_2 - S_2 + R_1 = 18$
$x_1, x_2, S_1, S_2 \ge 0$

Starting b.f.s $\Rightarrow$?

1	0
0	1
S_1	? R_1

$z_0 = 18M.\ (S_1, R_1) = (4, 18)$

Basic	x_1	x_2	S_1	S_2	R_1	R.H.S
z	3	-4	0	0	$-M$	0 $\rightarrow$!
S_1	1	1	1	0	0	4
R_1	2	3	0	-1	1	18

$$(R_1\text{-row})(.M) + \begin{matrix}\text{old}\\ z\text{-row}\end{matrix} = \begin{matrix}\text{New}\\ z\text{-row.}\end{matrix}$$

Alternative operation to preprocess

min $z = -3x_1 + 4x_2 + M.R_1$

$\min z = -3x_1 + 4x_2 + M.R_1$

s.t. $x_1 + x_2 + S_1 = 4$

$2x_1 + 3x_2 - S_2 + R_1 = 18$

$x_1, x_2, S_1, S_2 \geq 0$

$R_1 = 18 - 2x_1 - 3x_2 + S_2$

Substitute in z.

$z = -3x_1 + 4x_2 + M(18 - 2x_1 - 3x_2 + S_2)$

$z = 18M + (-2M - 3)x_1 + (4 - 3M)x_2 + MS_2$

$\boxed{z + (3+2M)x_1 + (3M-4)x_2 - MS_2 = 18M}$

	Basic	x_1	x_2 E.V.	S_1	S_2	R_1	R.H.S	min Ratio
min.	z	$3+2M$	$3M-4$	0	$-M$	0	$18M$	
L.V.	S_1	1	1	1	0	0	4	$4/1$ *
	R_1	2	3→0	0	−1	1	18	$18/3$
min +	z	$7-M$	0	$4-3M$	$-M$	0	$16+6M$	
	x_2	1	1	1	0	0	4	
	R_1	−1	0	−3	−1	1	6	

$\begin{pmatrix}\text{New} \\ x_2\text{-row}\end{pmatrix}(4-3M) + \begin{matrix}\text{Old} \\ z\text{-row}\end{matrix} = \begin{matrix}\text{New} \\ z\text{-row}\end{matrix}$

$4-3M$	$4-3M$	$4-3M$	0	0	$16-12M$
$3+2M$	$3M-4$	0	$-M$	0	$18M$
$7-M$	0	$4-3M$	$-M$	0	$16+6M$

* All the z-row values are nonpositive (−, or, 0). So, we found the optimal solution.

found the optimal solution.

$x_2^* = 4$ $\qquad$ $R_1^* = 6.$

$x_1^* = 0$, $S_1^* = 0$, $S_2^* = 0$

$z^* = 16 + 6M$

Example 1:

min $z= -3x_1+4x_2$

s.t. $x_1+x_2 \le 4$

$2x_1+3x_2 \ge 18$

$x_1, x_2 \ge 0$

$\Rightarrow$ $\min z = -3x_1 + 4x_2 + M \cdot R_1$

s.t. $x_1 + x_2 + S_1 = 4$

$2x_1 + 3x_2 - S_2 + R_1 = 18$

$x_1, x_2, S_1, S_2 \geq 0$

1st const $\Rightarrow 0 + 4 + 0 = 4$ ✓ It is satisfied.

2nd " $\Rightarrow 2.0 + 3.4 - 0 + 6 = 18$

$18 = 18$ ✓ " "

original const.

s.t. $x_1+x_2 \le 4$

$2x_1+3x_2 \ge 18$

$x_1, x_2 \ge 0$

$0 + 4 \overset{?}{\leq} 4 \Rightarrow$ It is sat.

$2.0 + 3.4 \geq 18 \Rightarrow$ It is not sat.

$(x_1, x_2) = (0, 4)$ is not a fes.

$\Rightarrow$ Since the optimal solution includes a positive artificial variable, the given LP is infeasible.

Example 2:

max $z= 5x_1+4x_2$

s.t. $x_1+2x_2 \ge 6$

$4x_1+2x_2 \le 8$

$x_1, x_2 \ge 0$

$\Rightarrow$ $\max z = 5x_1 + 4x_2 - MR_1$

$x_1 + 2x_2 - S_1 + R_1 = 6$

$4x_1 + 2x_2 + S_2 = 8$

$x_1, x_2, S_1, S_2, R_1 \geq 0.$

$$R_1 = 6 - x_1 - 2x_2 + S_1$$
$$z = 5x_1 + 4x_2 - M(6 - x_1 - 2x_2 + S_1)$$
$$z = (5-6M)x_1 + (4+2M)x_2 - 6M - MS_1$$

max

Basic	x_1	x_2 E.V	S_1	S_2	R_1	RHS	
z	$6M-5$	$-2M-4$	M	0	0	$-6M$	
L.V R_1	1	2	-1	0	1	6	$6/2$
S_2	4	2	0	1	0	8	$8/2$
z	$7M-3$	0	-2 E.V.	0	$2+M$	12	min Ratio
x_2	$1/2$	1	$-1/2$	0	$1/2$	3	—
S_2	3	0	1	1	-1	2	$2/1 \Rightarrow$ L.V.
z	$7M+3$	0	0	2	$-3/2$	16	
x_2	2	1	0	$1/2$	0	4	
S_1	3	0	1	1	-1	2	

Please repeat the computations and find my mistake.
And write the opt. sol. (Homework)

$$\underset{2(2+M)}{(4+2M)}(x_2) + (z\text{-row}) = \text{New } z\text{-row}$$

$2+M$	$4+2M$	$-2-M$	0	$2+M$	$12+6M$
$6M-5$	$-2M-4$	M	0	0	$-6M$

$7M-3 \quad 0 \quad -2 \quad 0 \quad 2+M \quad 12$

Free variables with upper bounds ($x_i \le u_i$)

If a free variable x_i has an upper bound u_i, that is

$\boxed{x_i \le u_i} \Rightarrow 0 \le \boxed{u_i - x_i} \Rightarrow x_i' \ge 0$ where $\boxed{x_i' = u_i - x_i}$.

$x_1 \le 4$

$x_1 = u_1' - x_1'$

Example 1: Transform the following LP to standart form.

max $z = x_1 + x_2$

s.t. $-x_1 + 2x_2 \le 8$

$x_1 + 2x_2 \le 10$

$\boxed{x_1 \le 8}$, $x_2 \ge 0$

x_1 can be $(-)$.

$x_1 \le 8$

$0 \le 8 - x_1$

$x_1' = 8 - x_1$, $\quad x_1' \ge 0$.

$x_1 = 8 - x_1'$

$8 - x_1' \le 8$

$-x_1' \le 0$

$x_1' \ge 0$

$$\max z = 8 - x_1' + x_2$$

s.t. $-8 + x_1' + 2x_2 \le 8$

$8 - x_1' + 2x_2 \le 10$

$x_1' \ge 0, \; x_2 \ge 0$.

$$\max z' = -x_1' + x_2 \quad \text{where} \quad z' = z - 8.$$

s.t. $x_1' + 2x_2 \le 16$

$-x_1' + 2x_2 \le 2$

$x_1' \ge 0 \quad x_2 \ge 0$.

max $z' = -x_1' + x_2$

s.t. $x_1' + 2x_2 + s_1 = 16$

$-x_1' + 2x_2 + s_2 = 2$

$x_1', x_2, s_1, s_2 \ge 0$.

St. form

$\vdots$

$x_1'^{*} = \dots \Rightarrow x_1 = ?$

$z = ?$

$$\max z = 10x_1 - 5x_2 - 2x_3$$
$$s.t. \quad -2x_1 + 3x_2 + 3x_3 \le 10$$
$$x_1 + x_2 - 5x_3 = 8$$
$$x_1 \ge -3 \Rightarrow x_1 \ (-).$$
$$-1 \le x_2 \le 1 \Rightarrow x_2 \ (-).$$
$$x_3 \text{ is free} \Rightarrow x_3 \ (-).$$

$-x_1 \le 3.$

a) Obtain the standart form of the given problem.

b) Write its starting *basic feasible solution*.

(1) $x_1 \geq -3 \Rightarrow \underbrace{x_1 + 3}_{x_1'} \geq 0 \Rightarrow x_1' = x_1 + 3$

$$\boxed{x_1 = x_1' - 3}$$

(2) $-1 \le x_2 \le 1 \Rightarrow 0 \le \underbrace{x_2 + 1}_{x_2'} \le 2.$

$$\boxed{x_2 = x_2' - 1}$$

$x_2' \geq 0, \quad x_2' \le 2.$

(3) x_3 is free $\Rightarrow \boxed{x_3 = x_3' - x_3''}$

$x_3', x_3'' \geq 0.$

$$\max z = 10x_1 - 5x_2 - 2x_3$$
$$s.t. \quad -2x_1 + 3x_2 + 3x_3 \le 10$$
$$x_1 + x_2 - 5x_3 = 8$$
$$x_1 \ge -3.$$
$$-1 \le x_2 \le 1$$
$$x_3 \text{ is free}$$

$\Rightarrow$

$$\max z = 10(x_1' - 3) - 5(x_2' - 1) - 2(x_3' - x_3'')$$
$$-2(x_1' - 3) + 3(x_2' - 1) + 3(x_3' - x_3'') \le 10$$
$$x_1' - 3 + x_2' - 1 - 5(x_3' - x_3'') = 8$$
$$x_2' \le 2.$$
$$x_1', x_2', x_3', x_3'' \geq 0.$$

$$\max z' = 10x_1' - 5x_2' - 2x_3' + 2x_3'' \qquad \text{where } z' = z + 25$$
$$-2x_1' + 3x_2' + 3x_3' - 3x_3'' \le 7$$
$$x_1' + x_2' - 5x_3' + 5x_3'' = 12$$

$$x_1', x_2', x_3', x_3'' \geq 0.$$

a) Std form of the given LP.

$$\max \; z' = 10x_1' - 5x_2' - 2x_3' + 2x_3'' - MR_1 \quad \text{where } z' = z + 25$$

$$-2x_1' + 3x_2' + 3x_3' - 3x_3'' + S_1 = 7$$

$$x_1' + x_2' - 5x_3' + 5x_3'' + R_1 = 12$$

$$x_1', x_2', x_3', x_3'', S_1, R_1 \geq 0.$$

b) Starting b.f.s = ? $(S_1, R_1) = (7, 12)$